

Jilei Lin

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Research Profile

I develop statistical methodology for heterogeneous, spatially dependent, and censored data, with applications in medicine, environmental health, spatial transcriptomics, and social science.

Core methods: Quantile regression; distributed inference; spatial analysis; dimension reduction; survival analysis; missing data analysis; longitudinal data analysis; transfer learning.

Education

2021–2026	Ph.D. in Statistics , The George Washington University, Washington, D.C. Dissertation: <i>Quantile Regression for Spatial Data and Censored Data</i> . Advisors: Huixia Judy Wang and Feifang Hu .
2019–2021	M.S. in Statistics , University of Illinois Urbana-Champaign, IL.
2015–2019	B.S. in Finance , <i>summa cum laude</i> , Kean University, Union, NJ.

Publications

Published and Accepted

- [1] **Smoothed Quantile Regression for Spatial Data.**
[Lin, J.](#), Wang, H., Wang, L., and Kim, M. **Accepted**; *Journal of Computational and Graphical Statistics (JCGS)*. Software: [jileil2/SQBiT](#).
- [2] **Hybrid Supervised-unsupervised Modeling for Post-hurricane Private Well Contamination Risk Score Using Empirical Validation and Community-Informed Assessment.**
[Lin, J.](#)[†], Zhang, J., Wei, E., Shea, K., Wang, H. J., Apanasovich, T. V., Liddie, J. M., Hernandez, E., Norford, C., and Hu, X. C. (2026). *GeoHealth*, 10(6), e2026GH001858. doi:10.1029/2026GH001858.
- [3] **Minimizing Post-shock Forecasting Error through Aggregation of Outside Information.**
[Lin, J.](#) and Eck, D. J. (2021). *International Journal of Forecasting*, 37(4), 1710–1727. doi:10.1016/j.ijforecast.2021.03.010.
- [4] **The Application of PROMETHEE Multi-criteria Decision Aid in Financial Decision Making: Case of Distress Prediction Models Evaluation.**
Mousavi, M. M. and [Lin, J.](#) (2020). *Expert Systems with Applications*. doi:10.1016/j.eswa.2020.113438.
- [5] **Identifying Opinion Leaders in Twitter during Social Events: Social Impact Theory from Influencer’s Perspective.**
[Lin, J.](#), Wang, Z., Huang, Y., and Chen, R. (2020). 2019 4th International Conference on Communication and Information Systems, 197–205. doi:10.1109/ICCIS49662.2019.00042.

[†]Corresponding author.

Revision-Stage Manuscripts

- [6] **Distributed Inference for Spatial Quantile Regression.**
[Lin, J.](#), Wang, H., and Wang, L. **Reject & Resubmit**; *Journal of the American Statistical Association: Theory and Methods (JASA T&M)*.
- [7] **Transfer Learning for Survival-based Clustering of Predictors with an Application to TP53 Mutation Annotation.**
Liu, X., [Lin, J.](#), Yan, H., Shi, H., Montellier, E., Chi, E. C., Hainaut, P., and Wang, W. **Major revision**; *Journal of the American Statistical Association: Applications and Case Studies (JASA A&CS)*.
- [8] **Censored Bent-line Quantile Regression with Application in a Study on Experimental Autoimmune Myasthenia Gravis.**
[Lin, J.](#), Wang, H., and Luo, J. **Major revision**; *Annals of Applied Statistics (AoAS)*.

Submitted Manuscripts

- [9] **Striational Antibody-Associated Myositis–Bridging the Gap between Thymoma and Myasthenia Gravis: A Systematic Review.**
Luo, J., [Lin, J.](#), Shymansky, J., and Wang, H. J. **Submitted**; *Nature Communications*. [Code](#).

Working Papers

- [10] **Shed Acetylcholine Receptors from Damaged Muscles Trigger Myasthenia Gravis in a Rodent Model.**
Lawson, B. S., Li, X., [Lin, J.](#), Kuzma, L. C., Pham, A., Wu, Y., Borbi, A., Xu, Q., Wang, H., Garden, O. A., and Luo, J. To be submitted.
- [11] **Quantile Regression for Spatial Survival Data.**
[Lin, J.](#) and Wang, H. J. Work in progress.
- [12] **Predicting Drinking Water Contaminants in Underserved Areas Using Transfer Learning: An Informative Subset-Based Multitask Quantile Deep Neural Network Approach.**
Liddie, J. L., [Lin, J.](#), Wei, E., Shea, K., Zhang, X., and Hu, X. Work in progress.

Software

SQBiT	R package for smoothed spatial quantile regression; companion to the JCGS paper. jileil2/SQBiT .
DISQ	R package for distributed spatial quantile regression; companion to the JASA T&M revision. jileil2/DISQ .
CBQR	R package for censored bent-line quantile regression; companion to the AoAS revision. jileil2/cbqr .
PFAS	Interactive Shiny application for national PFAS prediction and cost-sensitive screening. PFAS Risk Screening Tool .

Professional Experience

2025–2026	Research Assistant , Predicting Drinking Water Contaminants in Underserved Areas with Transfer Learning. Principal investigators: Xindi Hu and Xiaoke Zhang.
2025	Research Assistant , Data-Driven Solutions to Mitigate the Impact of Hurricane on Private Well, Drinking Water Quality, and Human Health. Principal investigator: Xindi Hu.
2024–2025	Research Assistant , Unlocking Complex Heterogeneity in Large-Scale Spatial-Temporal Data with Adaptive Quantile Learning. Principal investigator: Huixia Judy Wang.
2023–2024	Research Assistant , Antigen-Specific Immunosuppression of Myasthenia Gravis by CAR-Engineered Tregs. Principal investigator: Jie Luo.

Teaching Experience

2024	Teaching Assistant, Introduction to Mathematical Statistics II (STAT 4158), Introduction to Business and Economic Statistics (STAT 1051), Mathematical Statistics II (STAT 6202), GWU.
2023	Teaching Assistant, Regression Analysis (STAT 2118), Survival Analysis (STAT 6227), Introduction to Business and Economic Statistics (STAT 1051), GWU.
2022	Teaching Assistant, Data Analysis (STAT 6210), Mathematical Statistics I (STAT 6201), GWU.
2021	Teaching Assistant, Advanced Regression Analysis (STAT 527), University of Illinois Urbana-Champaign.
2020	Teaching Assistant, Applied Regression and Design (STAT 425), University of Illinois Urbana-Champaign.

Presentations

2026	StatConnect 2026. <i>Censored Bent-line Quantile Regression with Application in a Study on Experimental Autoimmune Myasthenia Gravis</i> .
2025	The Annual ASA/IMS Spring Research Conference on Statistics in Industry and Technology, New York, NY. <i>Distributed Inference for Spatial Quantile Regression</i> .
2025	StatConnect 2025, Vienna, VA. <i>Distributed Inference for Spatial Quantile Regression</i> .
2025	The Past, Present and Future of Statistics in the Era of AI, Washington, D.C. <i>Censored Bent-line Quantile Regression with Application in a Study on Experimental Autoimmune Myasthenia Gravis</i> .
2024	ICSA Applied Statistics Symposium, Nashville, TN. <i>Distributed Inference for Spatial Quantile Regression</i> .
2024	Statistics in the Age of AI, Washington, D.C. <i>Censored Bent-line Quantile Regression with Application in a Study on Experimental Autoimmune Myasthenia Gravis</i> .
2023	ICSA Applied Statistics Symposium, Ann Arbor, MI. <i>Smoothed Quantile Regression for Spatial Data</i> .
2018	INFORMS International Conference, Taipei, Taiwan. <i>The Application of PROMETHEE Multi-criteria Decision Aid in Financial Decision Making: Case of Distress Prediction Models Evaluation</i> .

Mentoring

2024–2025	Maxwell Beveridge, undergraduate student.
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Honors and Awards

2026	Minna Miran Kullback Memorial Prize for Leadership and Service, Department of Statistics, GWU.
2025	Ph.D. Student Poster Competition Award, StatConnect 2025, Vienna, VA.
2024	GW Statistics Cornfield Travel Award, Department of Statistics, GWU.

Professional Service

Referee	Computational Statistics & Data Analysis .
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Skills

Programming	R, Python, Bash, LaTeX.
HPC and Automation	SLURM job scripting, GPU/CPU cluster workflows, environment modules.
Reproducibility	Git/GitHub project management, version control, workflow organization.

References

Huixia Judy Wang	Professor and Chair, Department of Statistics, Rice University. jw322@rice.edu .
Feifang Hu	Professor and Chair, Department of Statistics, The George Washington University. feifang@gwu.edu .
Lily Wang	Professor, Department of Statistics, George Mason University. lwang41@gmu.edu .